

Tool-41 — Vendor Risk Assessment

MVP Tool | Team: Team: 7 Members

Sprint: Monday 14 April 2026 — Friday 9 May 2026 | Demo Day: 9 May 2026

[!] **READ THIS DOCUMENT COMPLETELY BEFORE WRITING A SINGLE LINE OF CODE.** This is your complete project guide — what to build, the full tech stack, exact folder structure, every task broken down day by day, GitHub submission steps, Demo Day checklist, and the 8 most common mistakes. Read it twice.

1. PROJECT OVERVIEW

You are building **Tool-41 — Vendor Risk Assessment** as your internship capstone project. This is a real-world, AI-powered web application built using an industry-standard tech stack. It will be the strongest item on your resume when you graduate.

Tool ID	Tool-41
Tool Name	Vendor Risk Assessment
Tool Type	MVP — Large Tool (full feature set)
Team	Team: 7 Members
Sprint	Monday 14 April – Friday 9 May 2026 (20 working days)
Demo Day	Friday 9 May 2026 — 8-minute live presentation
Backend Port	8080 Swagger: http://localhost:8080/swagger-ui.html
AI Port	5000 Health: http://localhost:5000/health
Frontend Port	80 Access: http://localhost
Repository	Link will be shared by your mentor before Day 1

2. TECH STACK

Use this exact stack. Do not change it. Before Day 2 read the getting-started docs for any technology you are unfamiliar with: spring.io/guides, console.groq.com/docs, docs.trychroma.com, react.dev, tailwindcss.com, docs.docker.com/compose.

Technology	What It Does in This Project
Java 17	Backend language. Download from adoptium.net
Spring Boot 3.x	Java framework — REST APIs, security, scheduling, email
PostgreSQL 15	Primary database. Run inside Docker
Redis 7	In-memory cache and AI response cache (15 min TTL)
Flyway	DB migrations — every schema change in a numbered SQL file
Spring Security + JWT	Login and ADMIN/MANAGER/VIEWER role-based access control
JavaMailSender	HTML email notifications for key business events
Swagger / OpenAPI 3.0	API docs auto-generated at <code>/swagger-ui.html</code>
Python 3.11	AI service language
Flask 3.x	Python web framework — AI microservice on port 5000
Groq API (LLaMA-3.3-70b)	AI model — free tier at console.groq.com , no credit card
ChromaDB	Vector database for the RAG pipeline
sentence-transformers	Converts text to vectors for ChromaDB
flask-limiter	Rate limiting — blocks IPs exceeding 30 req/min
React 18 + Vite	Frontend framework with fast build tool
Tailwind CSS	Utility CSS — style with class names
Axios	HTTP client for React to call the Java backend
Recharts	React chart library for analytics dashboard
Docker + Docker Compose	Runs all 5 services with one command: <code>docker-compose up</code>

3. FOLDER STRUCTURE — Create This on Day 1

Create all folders on Day 1 before writing any code. Use this exact layout throughout the sprint.

```
your-project-folder/
|-- backend/ <- Spring Boot project
| |-- src/main/java/com/internship/tool/
| | |-- controller/ <- REST endpoints (@RestController)
| | |-- service/ <- Business logic (@Service)
| | |-- repository/ <- DB queries (JpaRepository)
| | |-- entity/ <- JPA table models (@Entity)
| | |-- dto/ <- Request and Response POJOs
| | |-- config/ <- Security, Redis, Mail, Async
| | |-- exception/ <- Custom exceptions + @ControllerAdvice
| |-- scheduler/ <- @Scheduled reminder jobs
|-- src/main/resources/
| |-- db/migration/ <- V1__init.sql, V2__audit.sql ...
| |-- templates/ <- Thymeleaf HTML email templates
| |-- application.yml <- All config (env variable references)
|-- pom.xml
|
|-- ai-service/ <- Flask Python microservice
| |-- routes/ <- describe.py, recommend.py ...
| |-- services/ <- groq_client.py, chroma_client.py
| |-- prompts/ <- prompt template text files
| |-- app.py <- Flask entry point
| |-- Dockerfile
|-- requirements.txt
|
|-- frontend/ <- React + Vite frontend
| |-- src/
| | |-- components/ <- Reusable UI components
| | |-- pages/ <- List, Detail, Form, Login, Dashboard
| | |-- services/ <- Axios API call functions
| |-- App.jsx <- Root component + routing
|-- package.json
|
|-- docker-compose.yml <- Runs all 5 services together
|-- .env.example <- All required environment variables
+-- README.md
```

4. TEAM ROLES

Your mentor will coordinate teams and guide progress. Each role below has a clear set of responsibilities for the full sprint.

Role	Responsibilities
Java Developer 1	Spring Boot setup, Service layer, JWT auth, data seeder, README.md, opens Demo Day.
Java Developer 2	DB schema (Flyway), Repository layer, audit logging, email notifications, Docker Compose, performance.
Java Developer 3	Full React frontend — all pages, responsive design, AI panel, demo video recording.
AI Developer 1	Flask setup, primary prompts, /describe and /recommend endpoints, ChromaDB RAG pipeline.
AI Developer 2	GroqClient, /categorise and /generate-report, prompt tuning, AI response caching, AI docs.
AI Developer 3	Input sanitisation, rate limiting, AiServiceClient.java, OWASP ZAP testing, SECURITY.md.
Security Reviewer	Independent security testing each Friday, validates fixes, presents security on Demo Day.

Your Team

Member	Role	Contact
Member 1	Role will be confirmed by your mentor before Day 1	Contact your mentor for team WhatsApp group details
Member 2	Role will be confirmed by your mentor before Day 1	Contact your mentor for team WhatsApp group details
Member 3	Role will be confirmed by your mentor before Day 1	Contact your mentor for team WhatsApp group details
Member 4	Role will be confirmed by your mentor before Day 1	Contact your mentor for team WhatsApp group details
Member 5	Role will be confirmed by your mentor before Day 1	Contact your mentor for team WhatsApp group details
Member 6	Role will be confirmed by your mentor before Day 1	Contact your mentor for team WhatsApp group details
Member 7	Role will be confirmed by your mentor before Day 1	Contact your mentor for team WhatsApp group details

5. DAY-BY-DAY TASK BREAKDOWN (20 Working Days)

Each row is one task for one role on one specific day. Complete it fully before end of that working day. If blocked, tell your team immediately — do not stay stuck silently. Update your status daily: **Not Started** → **In Progress** → **Completed** → **Blocked**

[YEL] Week 1: Foundation		[ORN] Week 2: Core Features		[BLU] Week 3: Polish	[GRN] Week 4: Demo Prep	[PUR] Day 20: Demo Day
Day	Date	Track	Role	Task — Complete This Fully Today		Status
Day 1	Mon Apr 14	— Week 1 —				
		Java	Java Developer 1	Setup Spring Boot — folder structure, pom.xml with all dependencies, application.yml using \${ENV_VAR} placeholders for DB, Redis, mail, JWT		Not Started
		Java	Java Developer 2	Write Flyway V1 migration — create core table with all required columns, correct data types, indexes on key lookup fields		Not Started
		Java	Java Developer 3	Setup React with Vite — install Axios and Tailwind CSS, set VITE_API_URL in .env, create pages/, components/, services/ folders		Not Started
		AI	AI Developer 1	Setup Flask — create routes/, services/, prompts/ folders, write requirements.txt, write app.py entry point registering all blueprints		Not Started
		AI	AI Developer 2	Get Groq API key at console.groq.com, store in .env as GROQ_API_KEY, write test_groq.py to confirm the API call works		Not Started
		AI	AI Developer 3	Read tool spec fully. Create SECURITY.md — document 5 OWASP Top 10 risks with attack scenario and mitigation for each		Not Started
Day 2	Tue Apr 15	— Week 1 —				
		Java	Java Developer 1	Create JPA Entity — all fields with @Column, @CreatedDate/@LastModifiedDate, @EntityListeners(AuditingEntityListener.class)		Not Started
		Java	Java Developer 2	Create JPA Repository — extend JpaRepository, add custom @Query methods for search, filter by status, find by date range		Not Started
		Java	Java Developer 3	Build list view page — table with all columns, Axios GET on mount, loading skeleton while fetching, empty state message		Not Started
		AI	AI Developer 1	Write primary prompt template for /describe — test with 5 real inputs, refine until outputs are consistently professional		Not Started
		AI	AI Developer 2	Implement GroqClient — API call, JSON parsing, 3-retry with backoff, error logging. Save as services/groq_client.py		Not Started
		AI	AI Developer 3	Document 5 security threats specific to this tool in SECURITY.md — attack vector, damage potential, mitigation plan for each		Not Started

Day 3	Wed Apr 16	— Week 1 —			
		Java	Java Developer 1	Implement Service layer — all business logic, input validation, custom exception classes for each error type	Not Started
		Java	Java Developer 2	Write Flyway V2 — create audit_log table, composite index on (entity_type, entity_id), additional indexes on key query columns	Not Started
		Java	Java Developer 3	Build create/edit form — controlled inputs for all fields, client-side validation with inline error messages, submit handler	Not Started
		AI	AI Developer 1	Build POST /describe — validate input, load prompt template, call Groq, return structured JSON with generated_at timestamp	Not Started
		AI	AI Developer 2	Build POST /categorise — classify input into predefined categories, return {category, confidence: 0.0-1.0, reasoning}	Not Started
		AI	AI Developer 3	Implement input sanitisation middleware — strip HTML, detect prompt injection patterns, return 400 with clear error message	Not Started
Day 4	Thu Apr 17	— Week 1 —			
		Java	Java Developer 1	Build REST Controller — GET /all paginated, GET /{id} with 404, POST /create with @Valid, all correct HTTP status codes	Not Started
		Java	Java Developer 2	Build REST Controller — PUT /{id}, DELETE (soft delete), GET /search?q=, GET /stats for dashboard KPIs	Not Started
		Java	Java Developer 3	Connect list page to GET /all API — handle Spring Page response, implement pagination controls, sort by column headers	Not Started
		AI	AI Developer 1	Build POST /recommend — return 3 actionable recommendations as JSON array, each with action_type, description, priority	Not Started
		AI	AI Developer 2	Setup ChromaDB — init persistent collection, write test verifying embed + query returns correct result	Not Started
		AI	AI Developer 3	Add flask-limiter — 30 req/min default, 10 req/min on /generate-report, return 429 with retry_after on breach	Not Started
Day 5	Fri Apr 18	— Week 1 —			
		Java	Java Developer 1	Implement JWT auth — JwtUtil (generate/validate/extract), JwtAuthFilter, SecurityConfig (permit /auth/**, require all others)	Not Started
		Java	Java Developer 2	Build AuthController — POST /login (validate, return JWT), POST /register (hash password, VIEWER role), POST /refresh	Not Started
		Java	Java Developer 3	Build login page — form, store JWT, AuthContext with isAuthenticated, ProtectedRoute redirects if not logged in	Not Started
		AI	AI Developer 1	Build RAG pipeline — load docs, chunk (500 chars, 50 overlap), embed with sentence-transformers, store all in ChromaDB	Not Started

		AI	AI Developer 2	Build POST /query — embed question, retrieve top-3 ChromaDB chunks, inject as context, call Groq, return answer + sources	Not Started
		AI	AI Developer 3	Week 1 security test — test all endpoints for empty input, SQL injection patterns, prompt injection. Document pass/fail in SECURITY.md	Not Started
Day 6	Mon Apr 21	— Week 2 —			
		Java	Java Developer 1	Add Redis caching — @Cacheable on GETs with 10 min TTL, @CacheEvict on all writes. Test cache hit vs miss timing	Not Started
		Java	Java Developer 2	Implement RBAC — ADMIN/MANAGER/VIEWER roles, @PreAuthorize on endpoints, seed roles via Flyway V3 migration	Not Started
		Java	Java Developer 3	Build dashboard — 4 KPI cards from GET /stats, Recharts BarChart showing data by category or status	Not Started
		AI	AI Developer 1	Build POST /generate-report — produce {title, executive_summary, overview, top_items, recommendations} as JSON	Not Started
		AI	AI Developer 2	Prompt tuning — run all prompts against 10 real inputs, score accuracy and format, rewrite any below 7/10	Not Started
		AI	AI Developer 3	Write AiServiceClient.java — RestTemplate calls to all Flask endpoints, 10s timeout, graceful null return on error	Not Started
Day 7	Tue Apr 22	— Week 2 —			
		Java	Java Developer 1	Email notifications — JavaMailSender, Thymeleaf HTML templates, send on create and overdue events	Not Started
		Java	Java Developer 2	@Scheduled cron jobs — daily reminder for overdue items, 7-day advance deadline alert, weekly summary	Not Started
		Java	Java Developer 3	Detail page — all fields, colour-coded score/status badge, Edit/Delete buttons, AI Analysis card section	Not Started
		AI	AI Developer 1	Integrate AiServiceClient into Service — call AI on create @Async, attach result to entity, handle null gracefully	Not Started
		AI	AI Developer 2	Build GET /health — model name, avg response time (last 10), ChromaDB doc count, uptime, cache stats	Not Started
		AI	AI Developer 3	Run OWASP ZAP baseline scan — export report, categorise findings by severity, plan remediation for all Medium+	Not Started
Day 8	Wed Apr 23	— Week 2 —			
		Java	Java Developer 1	@ControllerAdvice — catch ResourceNotFoundException (404), ValidationException (400), generic Exception (500), consistent JSON body	Not Started
		Java	Java Developer 2	Audit logging via Spring AOP — @Around on service CUD methods, save old/new JSON to audit_log table	Not Started
		Java	Java Developer 3	AI panel in UI — 'Ask AI' button, loading spinner, formatted AI response card, Retry on error	Not Started

		AI	AI Developer 1	Add SSE streaming to /generate-report — stream tokens to frontend, React reads with EventSource	Not Started
		AI	AI Developer 2	Redis AI cache — SHA256 cache key, 15 min TTL, increment hit/miss counters, skip cache for fresh requests	Not Started
		AI	AI Developer 3	Fix all ZAP findings — add security headers (X-Content-Type-Options, X-Frame-Options), re-scan to confirm resolved	Not Started
Day 9	Thu Apr 24	— Week 2 —			
		Java	Java Developer 1	File attachments — POST /upload, GET /files/{id}, validate type and size (< 10 MB), store with UUID filename	Not Started
		Java	Java Developer 2	Add pagination params to all list endpoints — page, size, sortBy, sortDir. Add GET /export CSV endpoint	Not Started
		Java	Java Developer 3	Search and filter bar — debounced 300ms text search, status dropdown, date range picker, URL query params	Not Started
		AI	AI Developer 1	Build POST /analyse-document — accept text, identify key insights and risks, return structured findings array	Not Started
		AI	AI Developer 2	Add meta object to all responses — {confidence: 0.0-1.0, model_used, tokens_used, response_time_ms, cached: bool}	Not Started
		AI	AI Developer 3	PII audit — verify no personal data in prompts or logs. Document findings in SECURITY.md	Not Started
Day 10	Fri Apr 25	— Week 2 —			
		Java	Java Developer 1	Write 10 JUnit 5 unit tests for Service — Mockito mocks, happy path and error cases for every public method	Not Started
		Java	Java Developer 2	Write MockMvc integration tests — test every endpoint for correct HTTP status codes and response body structure	Not Started
		Java	Java Developer 3	React export — CSV download button, streaming report display using EventSource, analytics page with Recharts	Not Started
		AI	AI Developer 1	Write 10 pytest unit tests — mock Groq responses, test each endpoint for format, error handling, edge cases	Not Started
		AI	AI Developer 2	Week 2 AI quality review — 10 fresh inputs per endpoint, score accuracy (target avg >= 4/5), rewrite failing prompts	Not Started
		AI	AI Developer 3	Week 2 security sign-off — JWT enforcement, rate limiting, injection rejection all verified. Sign off in SECURITY.md	Not Started
Day 11	Mon Apr 28	— Week 3 —			
		Java	Java Developer 1	Additional tests to reach 80% coverage — @DataJpaTest for Repository, AuthController, JwtUtil, AuditService	Not Started
		Java	Java Developer 2	@SpringBootTest integration test — full CRUD flow with Testcontainers (real PostgreSQL + Redis containers)	Not Started
		Java	Java Developer 3	Analytics page — BarChart by category, LineChart over time (6 months), PieChart by status, period selector	Not Started
		AI	AI Developer 1	Build POST /batch-process — accept up to 20 items, process each with 100ms delay, return results array	Not Started

		AI	AI Developer 2	Async job processing for /generate-report — return job_id immediately, background thread, webhook on complete	Not Started
		AI	AI Developer 3	Full OWASP ZAP active scan — fix all Critical and High findings today, document Medium as accepted or planned	Not Started
Day 12	Tue Apr 29	— Week 3 —			
		Java	Java Developer 1	Swagger/OpenAPI — @Operation and @ApiResponse on every controller method, document all DTOs with examples	Not Started
		Java	Java Developer 2	docker-compose.yml — all 5 services with healthchecks, depends_on, env from .env, test with docker-compose up --build	Not Started
		Java	Java Developer 3	UX polish — loading skeletons, empty state illustrations, React Error Boundary on all pages	Not Started
		AI	AI Developer 1	Write ai-service/README.md — prerequisites, setup steps, env vars, run instructions, full endpoint API reference	Not Started
		AI	AI Developer 2	Performance benchmark — p50/p95/p99 for all endpoints, 50 requests each. Optimise any endpoint exceeding target	Not Started
		AI	AI Developer 3	Fix all remaining ZAP findings — flask-talisman for security headers, re-scan confirms zero Critical/High remaining	Not Started
Day 13	Wed Apr 30	— Week 3 —			
		Java	Java Developer 1	Full system test — docker-compose down -v then up --build, walk every feature end-to-end, document all bugs	Not Started
		Java	Java Developer 2	Fix all bugs from system test — P1 crashes and P2 wrong data fixed today, zero P1/P2 allowed after Day 13	Not Started
		Java	Java Developer 3	Responsive design — test 375px / 768px / 1280px, fix all layout breakages with Tailwind responsive prefixes	Not Started
		AI	AI Developer 1	Optimise slow endpoints — pre-load sentence-transformers model at startup, reduce prompt length, verify Redis cache	Not Started
		AI	AI Developer 2	AI fallback — on Groq timeout/error, return pre-written template response with {is_fallback: true} in meta	Not Started
		AI	AI Developer 3	Full stack security test — 401 without token, 403 wrong role, XSS in input field, 429 after rate limit, all verified	Not Started
Day 14	Thu May 1	— Week 3 —			
		Java	Java Developer 1	Data seeder — DataLoader seeds 30 realistic demo records on startup, covering all statuses and score ranges	Not Started
		Java	Java Developer 2	Performance — EXPLAIN ANALYZE slow queries, add missing indexes, fix N+1 with @EntityGraph or JOIN FETCH	Not Started
		Java	Java Developer 3	UI brand polish — consistent primary colour (#1B4F8A), Arial font, 8px grid spacing, 44px min touch targets	Not Started
		AI	AI Developer 1	Seed ChromaDB — ingest 10 domain knowledge documents covering key topics relevant to this tool	Not Started
		AI	AI Developer 2	Final prompt QA — run all prompts against 30 seeded demo records, verify all outputs are demo-ready	Not Started

		AI	AI Developer 3	Final SECURITY.md — executive summary, all threats, tests conducted, findings fixed, residual risks, team sign-off	Not Started
Day 15	Fri May 2	— Week 3 —			
		Java	Java Developer 1	Team code review — all PRs reviewed, no hardcoded secrets, no TODO comments, coding standards enforced	Not Started
		Java	Java Developer 2	Final merge — clean history, docker-compose up --build passes, all tests pass, tag release/v1.0	Not Started
		Java	Java Developer 3	Record 2-min demo video — screen record with voice narrating every key feature, save as demo_video.mp4	Not Started
		AI	AI Developer 1	Final AI QA — all 6 endpoints tested with demo data, every output professional and demo-ready	Not Started
		AI	AI Developer 2	Package AI — Dockerfile builds and runs cleanly, requirements.txt exact versions, .env.example complete	Not Started
		AI	AI Developer 3	Final security checklist — tick every item, all 6 members sign off, commit SECURITY.md final version	Not Started
Day 16	Mon May 5	— Week 4 —			
		Java	Java Developer 1	Write README.md — overview, ASCII architecture diagram, prerequisites, setup steps, .env.example reference table	Not Started
		Java	Java Developer 2	Final bug fix — zero P1/P2 bugs after today, P3 documented as known issues in README	Not Started
		Java	Java Developer 3	Prepare 5 demo scenarios — exact steps and expected output for each, walk through on live system and confirm	Not Started
		AI	AI Developer 1	Write AI demo script — exact inputs, expected outputs, and what to say for each AI interaction in the 8-min demo	Not Started
		AI	AI Developer 2	Final performance verify — all endpoints within targets, Redis cache confirmed working, fallback confirmed working	Not Started
		AI	AI Developer 3	Prepare security talking points — 4 bullet points covering JWT, rate limiting, input sanitisation, ZAP results	Not Started
Day 17	Tue May 6	— Week 4 —			
		Java	Java Developer 1	Demo Rehearsal 1 — full team, stopwatch, complete 8-min presentation, note all issues to fix	Not Started
		Java	Java Developer 2	Fix all rehearsal issues — data, transitions, bugs. Re-test every scenario after fixes	Not Started
		Java	Java Developer 3	Prepare 3 slides — Slide 1: problem, Slide 2: architecture, Slide 3: demo flow. Keep slides to 1 min total	Not Started
		AI	AI Developer 1	AI dry run on demo machine — all 6 endpoints live with Groq API, record all response times	Not Started
		AI	AI Developer 2	Backup deck — screenshot every AI output from dry run in case Groq API fails during live demo	Not Started
		AI	AI Developer 3	Individual practice — each member presents their section solo without notes, identify and fix gaps	Not Started

Day 18	Wed May 7	— Week 4 —			
		Java	Java Developer 1	Submit GitHub — public, no secrets, one commit per day since Day 1, share link with mentor	Not Started
		Java	Java Developer 2	Fresh machine test — clone repo, create .env, docker-compose up --build, full stack healthy in under 3 min	Not Started
		Java	Java Developer 3	5 portfolio screenshots at 1920x1080 — dashboard, detail, AI panel, analytics, report streaming	Not Started
		AI	AI Developer 1	Submit AI service GitHub link — Dockerfile builds cleanly, README complete, share link with mentor	Not Started
		AI	AI Developer 2	Create 1-page AI summary card — 6 endpoints, tech stack, GitHub link. Print 2 copies for Demo Day	Not Started
		AI	AI Developer 3	Submit final SECURITY.md — professional, complete, accurate. Print 1 copy for Demo Day	Not Started
Day 19	Thu May 8	— Week 4 —			
		Java	Java Developer 1	Rehearsal 2 — full team, live system, 8 min target. Every member answers 5 key questions without notes	Not Started
		Java	Java Developer 2	Final hotfix window — P1/P2 fixes only, no new features, re-test after each fix	Not Started
		Java	Java Developer 3	docker-compose down -v then up — fresh seeded state, all 30 records visible, all 5 scenarios confirmed	Not Started
		AI	AI Developer 1	Groq API check — key active, credits sufficient, test call to all 6 endpoints confirmed working	Not Started
		AI	AI Developer 2	Print AI talking points card — Groq, RAG, ChromaDB, prompt engineering each in 2 plain-English sentences	Not Started
		AI	AI Developer 3	Final confidence check — every member presents their 90-sec section solo, resolve any remaining uncertainty	Not Started
Day 20	Fri May 9	— Demo —			
		Java	Java Developer 1	DEMO — Open: problem statement (1 sentence), architecture slide, launch live tool. Create record, watch AI respond	Not Started
		Java	Java Developer 2	DEMO — Show audit log, email notification, update status, demonstrate search and filter, CSV export	Not Started
		Java	Java Developer 3	DEMO — Dashboard KPIs, analytics charts, responsive design on mobile window, final UI walkthrough	Not Started
		AI	AI Developer 1	DEMO — AI Recommend (read aloud), Generate Report (show streaming), RAG query. Explain what AI is doing	Not Started
		AI	AI Developer 2	DEMO — Explain stack: Flask + Groq + ChromaDB RAG in 60 seconds, show /health endpoint response	Not Started
		AI	AI Developer 3	DEMO — Security: 401 without token, 400 on injection attempt, reference SECURITY.md, state findings fixed	Not Started

6. GITHUB — Push Code Every Working Day

Push your code at the end of every working day. Your mentor will share the repository details before Day 1.

Step	Action	Instructions
1	Create Account	Go to github.com, click Sign Up. Use your personal email. Choose a professional username.
2	Create Repository	Your mentor will share the exact repository name and URL. Set Visibility to Public. Add README.
3	Install Git	Download from git-scm.com. In terminal run: git --version to confirm version 2.x or higher.
4	Configure Git	Run: git config --global user.name 'Your Name' and git config --global user.email 'you@email.com'
5	Clone Repo	Run: git clone [repository URL shared by mentor] then cd into the project folder
6	Add .gitignore	On Day 1, before any commit: create .gitignore and add: .env target/ node_modules/ __pycache__/ chroma_data/
7	Daily Push	End of every working day: git add . then git commit -m 'Day X — what you did' then git push origin main
8	Verify Online	After every push: open the repository URL in your browser and confirm the latest files are visible.

Commit message format every day: Day X — brief description of what you completed

Example: Day 3 — Implemented Service layer with validation and custom exceptions

7. DEMO DAY CHECKLIST

Every item must be complete before your 8-minute Demo Day presentation on 9 May 2026.

Deliverable	What Must Be Submitted or Demonstrated	Done?
GitHub Repo	Public repo (link from mentor). At least 1 commit per working day. No secrets committed.	☐
README.md	Complete setup instructions. ASCII architecture diagram. Prerequisites. Step-by-step setup.	☐
docker-compose	All services defined. Full stack starts with docker-compose up. Tested on a clean machine.	☐
Flyway	All schema changes in numbered files (V1__, V2__, ...). No manual SQL outside migrations.	☐
JUnit Tests	Minimum 10 tests, all passing. JaCoCo coverage $\geq 80\%$.	☐
Pytest Tests	Minimum 8 tests, all passing. Groq API mocked — tests run without live network access.	☐
Swagger UI	All endpoints documented at /swagger-ui.html with request/response examples.	☐
AI Service	Flask app with working endpoints. Dockerfile builds. README covers setup.	☐
SECURITY.md	Threat model, tests conducted, findings fixed, residual risks, team sign-off.	☐
Demo Data	30 realistic seeded records on docker-compose up, covering all statuses and ranges.	☐
Demo Video	demo_video.mp4: 90-120 seconds, all key features shown with clear voice narration.	☐

8. DEMO DAY — Friday 9 May 2026

You have **8 minutes**. The tool must be running live on your laptop. No slides-only presentations. Real features, real data, live AI calls.

Segment	Time	What to Do and Say
Opening	1 min	State the problem in one sentence. Show architecture slide. Say 'Let me show you the live tool.'
CRUD Demo	2 min	Create a record live, watch AI description appear. Update status. Show audit log. Mention email notification.
AI Features	2 min	Click AI Recommend (read aloud). Generate Report (show SSE streaming). RAG query demo.
UI + Analytics	1 min	Dashboard KPIs. Analytics charts. CSV export. Responsive design on mobile-sized window.
Security	1 min	API call without JWT (show 401). Injection attempt (show 400). Reference SECURITY.md. State findings fixed.
Q&A;	1 min	Every member answers: What does it do? What AI? What is RAG? What security measures?

9. COMMON MISTAKES — Read This Now

These 8 mistakes are made every sprint. Avoiding them saves days of wasted work.

Mistake	Why It Happens and How to Avoid It
[X] Hardcoding secrets in config files	Use <code>\${ENV_VAR}</code> in <code>application.yml</code> and <code>os.getenv()</code> in Python. Actual values go only in <code>.env</code> (which is in <code>.gitignore</code>). If you commit a secret to GitHub, rotate it immediately — deleting the file is not enough.
[X] Committing <code>.env</code> to GitHub	Add <code>.env</code> to <code>.gitignore</code> on Day 1 before any commit. Run <code>git status</code> before every git commit to verify. GitHub scans for exposed keys and will warn you, but by then it is already public.
[X] Not wrapping Groq calls in try-except	The Groq free tier has rate limits and occasional outages. Every call must be in a try-except block. On failure: log the error, return the fallback template with <code>{is_fallback: true}</code> . Never return HTTP 500 because the AI is unavailable.
[X] Not setting up docker-compose until Week 3	Set it up on Day 5. The most common Demo Day failure is 'works on my laptop but not the demo machine.' Test the full docker-compose stack every day from Day 5 onwards. Discover integration problems early.
[X] Adding features in Week 4	Week 4 is for demo preparation only. Feature freeze after Day 15. Every hour building features in Week 4 is an hour not spent making the demo flawless. New ideas go in GitHub Issues labelled post-sprint.
[X] Empty database on Demo Day	Implement the DataLoader seeder on Day 14. On Demo Day: docker-compose down -v then docker-compose up to reset to a clean seeded state. An empty list on Demo Day undermines the entire presentation.
[X] Not reading Groq docs before writing AI code	Spend 30 minutes at console.groq.com/docs before Day 2. Key parameters: model name, temperature (0.3 for factual, 0.7 for creative), <code>max_tokens</code> , and the messages array structure. Misunderstanding temperature alone can waste an entire day.
[X] Skipping the demo dry run	Every team that skips the rehearsal on Day 17 regrets it. Problems are always found in rehearsal that you never noticed before. Do two full rehearsals (Day 17 and Day 19). Use a stopwatch. The time limit is firm.